

Building a Khmer Traffic Sign Dataset for Automated Detection and Classification

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Abstract

Traffic Sign Detection and Classification have become a core feature of computer vision, contributing for intelligent transportation systems, traffic monitoring and driver assistance technologies. However, the availability of public datasets illustrating Khmer traffic signs remains limited, making it challenging to develop and evaluate reliable detection and classification models for Cambodia. To overcome the lack of the publicly available Khmer datasets, this research introduces a custom Khmer traffic sign datasets collected and annotated the capture image from various locations in Cambodia. The datasets contain an image captured under multiple real-world conditions including different lighting, different angles, distance, background and covers 42 classes commonly found in Cambodia. To evaluate the dataset, YOLOv8 and YOLOv10 models were trained and compared. YOLOv8 achieved the best performance with an accuracy of 93% on the primary dataset and was integrated into a bilingual web-based application supporting both Khmer and English languages for automated traffic sign detection and classification. This system demonstrates the practical application of deep learning-based Khmer traffic sign recognition and provides a foundation for future research and intelligent transportation applications in Cambodia.

Keywords: *Khmer Traffic Sign, Khmer Traffic Sign Detection and Classification (KTSDC), Computer Vision, Deep Learning*

1 Introduction

Traffic Signs play a significance role in modern transportation systems by providing an importance information, guiding driver behavior and enhancing road safety [1][2]. With the rapid development of computer vision and deep learning techniques, automated traffic sign detection

and classification have become main research areas for intelligent transportation systems and driver assistance technologies.

Developing accurate and reliable traffic sign detection and classification models require a large-scale dataset. Existing datasets, including German Traffic Sign Recognition Benchmark (GTSRB) [3], Mapillary Traffic Sign Dataset (MTSD) [1], Tsinghua-Tencent 100K (TT100K) [4], provide comprehensive coverage for their regions. However, they are limited for specific countries and do not effectively represent the traffic sign used in Cambodia [1], [3], [4].

In Cambodia, the availability of publicly accessed Khmer traffic sign datasets remains limited [5]. Existing research on Cambodian traffic signs focuses only on seven classes [5], which do not fully represent the wide range of usually found in Cambodia roadways. Therefore, this limitation highlights a significant research gap in the development of a complete KTSDC system.

To address this research gap, this study proposes a KTSDC system based on a custom traffic sign dataset that was collected and annotated from various locations across Cambodia. The proposed collection consists of 42 classes of traffic sign types collected under various real-world conditions, including different lighting, different angles, distances, and backgrounds. Furthermore, YOLOv8 and YOLOv10 were trained and evaluated to find out the effectiveness for Khmer traffic sign detection and classification.

2 Dataset

In this section, we will describe the custom Khmer traffic sign dataset produced in this project. The dataset was manually collected and annotated for traffic sign detection and classification in Cambodia road environments.

2.1 Dataset Overview

This paper introduces custom Khmer traffic sign dataset designed for traffic sign detection and classification in Cambodia road conditions. The proposed dataset was collected and annotated by our team, which consisted of 42 classes that consisting several categories including regulation, warning, required, and informational signals. The collected images demonstrate various real-world environments with differences in lighting, angles, distances and backgrounds. Figure 1 demonstrates the complete set of traffic sign classes included in the proposed dataset. In the next section will detail the data collection, annotation and data statistics.



Figure 1. The 42 traffic sign classes included in the proposed dataset

2.2 Data Collection

The traffic sign images were manually collected by our team from several sites across Cambodia using a smartphone camera. Data collecting was conducted on a few different road types, which include urban streets and rural streets. Images were taken under different real-world conditions, such as different lighting conditions, viewing angles, object distances, and background environments, to boost dataset diversity. After data gathering, all the images were carefully reviewed to remove duplicate images, images without traffic signs, and images that contain poor visibility that prevented clear identification of the traffic signs. This process ensured that only the quality images were left for data annotation process.

2.3 Data Annotation

Following data collection, all the images were manually annotated using the CVAT annotation tool. Each traffic sign was allocated a rectangular bounding box and provided with a class label according to one of the 42 predefined traffic sign

classes. The annotations were exported in the YOLO format, containing the class identifier and normalized bounding box coordinates for each signs. To ensure annotations quality, all annotations have been manually validated by our team, and incorrect class labels or inaccurate bounding boxes were fixed before finish datasets.

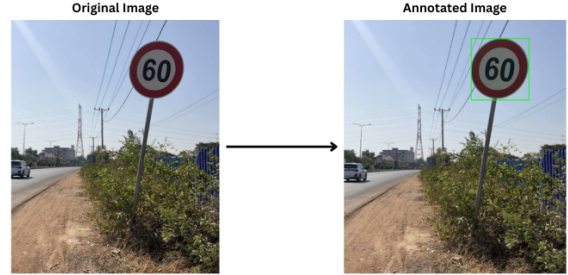


Figure 2. Example of the traffic sign annotation process. Original image. Annotated image with bounding box and class label generated using CVAT

2.4 Data Statistics

The proposed dataset consisted of 31667 images illustrating 42 classes and 31734 for annotated traffic signs, while 123 images are the images contain backgrounds images without traffic signs annotation. The number of annotated images exceeds the number of images because some images contain multiple annotated traffic signs. The distribution of annotated across the 42 classes demonstrated in Figure 3, displaying the variance in each class represent within the dataset. To simplify model training, validation and testing. The dataset was divided into training, validation, and testing subsets using a [70:20:10] ratio. The training subset was used to enhance the model parameters, while validation was utilized to check model performance throughout training. The testing subset designed for the final evaluation of the train models. The number of propotion of images assigned to each subset are summarizezed in Table 1.

3 Methodology

The following section discusses the techniques utilized to develop and create the Khmer traffic sign detection and classification approach. The overall research process and experimental procedures are describe in the following section.

Table 1. Distribution of Images and Annotations in the Proposed Dataset

Dataset Subset	Total	Training	Validation	Testing
Images	31667	22148	6318	3201
Labels	31544	22064	6292	3188
Bounding Box	31734	22195	6335	3204

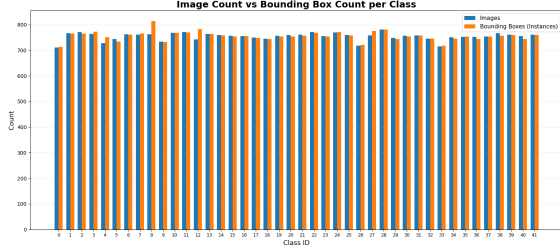


Figure 3. Class distribution of images and annotated traffic signs across the 42 classes.

3.1 Methodology Workflow

The proposed methodology establishes a systematic workflow for Khmer traffic sign detection and classification. The process begins with data preparation and data preprocessing for the annotated traffic sign dataset, followed by the model development and training under experimental set up. After training, the models will evaluated on the test dataset using standard object detection metrics. The obtained results are subsequently compared to identify model that achieves performance for Khmer dataset.

3.2 Data Preprocessing

3.3 Model Development

3.4 Model Training

3.5 Evaluation

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Table 2. Units for Magnetic Properties

Subset	Total	Training	Validation	Testing
Images	31667	22148		
Labels	31544	22064		
Bouding Box	32036	22400		

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Format and save your graphics using a suitable graphics processing program that will allow you to create the images as PostScript (.PS), Encapsulated PostScript (.EPS), Tagged Image File Format (.TIFF), Portable Document Format (.PDF), Portable Network Graphics (.PNG), or Metapost (.MPS), sizes them, and adjusts the resolution settings. When submitting your final paper, your graphics should all be submitted individually in one of these formats along with the manuscript.

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Figure 4. Sample Image

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Acknowledgment

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